A Periodic Table of the Finite Simple Groups

Every finite simple group appears exactly once, in its simplest form. Columns are grouped by rank; the vertical axis is the field size q.

TYPE

🟧 Cyclic groups · 🟪 Alternating groups · 🟦 Chevalley groups · 🟩 Steinberg groups · 🟨 Suzuki and Ree groups · 🟪 Sporadic groups

HOW TO READ A CELL

$\cong A_2(2)$ — other descriptions of the same group

$A_1(7)$ — the group

$7(7^2-1)/2$ — prime factorisation of its order

AN EMPTY CELL

the group is not defined there, or is not simple, or it is isomorphic to one printed elsewhere — every group appears only under its simplest name:

$A_2(1) \to C_3$
$A_1(4), A_1(5) \to A_5$ [$A_4(1)$]
$A_1(9) \to A_6$ [$A_5(1)$]
$A_2(2) \to A_1(7)$
$A_3(2) \to A_8$ [$A_7(1)$]
${}^2A_3(2) \to B_2(3)$
$C_n(q)$, q even $\to B_n(q)$
$B_2(2)' \to A_6$ [$A_5(1)$]
$G_2(2)' \to {}^2A_2(3)$
${}^2G_2(3)' \to A_1(8)$

FAMILIES

$A_n(q) = L_{n+1}(q)$ — Linear
$B_n(q) = O_{2n+1}(q)$ — Orthogonal (odd)
$C_n(q) = S_{2n}(q)$ — Symplectic
$D_n(q) = O^+_{2n}(q)$ — Orthogonal (plus)
${}^2A_n(q) = U_{n+1}(q)$ — Unitary
${}^2D_n(q) = O^-_{2n}(q)$ — Orthogonal (minus)
$G_2(q)$ — Exceptional G_2
$F_4(q)$ — Exceptional F_4
$E_6(q)$ — Exceptional E_6
$E_7(q)$ — Exceptional E_7
$E_8(q)$ — Exceptional E_8
${}^3D_4(q)$ — Triality
${}^2E_6(q)$ — Twisted E_6
${}^2B_2(q)$ — Suzuki
${}^2G_2(q)$ — Ree (small)
${}^2F_4(q)$ — Ree (large)

	cyclic	rank 1	rank 2						rank 3
	C_p	A_1	A_2	B_2	${}^2A_2$	G_2	${}^2B_2$	${}^2G_2$	A_3
1 “F_1”									
2 prime	C_2 2								
3 prime	$\cong A_2(1)$ C_3 3		$A_2(3)$ $2^4 \cdot 3^3 \cdot 13$	$\cong {}^2A_3(2)$ $B_2(3)$ $2^6 \cdot 3^4 \cdot 5$	$\cong G_2(2)'$ ${}^2A_2(3)$ $2^5 \cdot 3^3 \cdot 7$	$G_2(3)$ $2^6 \cdot 3^6 \cdot 7 \cdot 13$			$A_3(3)$ $2^7 \cdot 3^6 \cdot 5 \cdot 13$
4 2^2			$A_2(4)$ $2^6 \cdot 3^2 \cdot 5 \cdot 7$	$\cong C_2(4)$ $B_2(4)$ $2^8 \cdot 3^2 \cdot 5^2 \cdot 17$	${}^2A_2(4)$ $2^6 \cdot 3 \cdot 5^2 \cdot 13$	$G_2(4)$ $2^{12} \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 13$			$A_3(4)$ $2^{12} \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 17$
5 prime	C_5 5		$A_2(5)$ $2^5 \cdot 3 \cdot 5^3 \cdot 31$	$B_2(5)$ $2^6 \cdot 3^2 \cdot 5^4 \cdot 13$	${}^2A_2(5)$ $2^4 \cdot 3^2 \cdot 5^3 \cdot 7$	$G_2(5)$ $2^6 \cdot 3^3 \cdot 5^6 \cdot 7 \cdot 31$			$A_3(5)$ $2^7 \cdot 3^2 \cdot 5^6 \cdot 13 \cdot 31$
7 prime	C_7 7	$\cong A_2(2)$ $A_1(7)$ $2^3 \cdot 3 \cdot 7$	$A_2(7)$ $2^5 \cdot 3^2 \cdot 7^3 \cdot 19$	$B_2(7)$ $2^8 \cdot 3^2 \cdot 5^2 \cdot 7^4$	${}^2A_2(7)$ $2^7 \cdot 3 \cdot 7^3 \cdot 43$	$G_2(7)$ $2^8 \cdot 3^3 \cdot 7^6 \cdot 19 \cdot 43$			$A_3(7)$ $2^9 \cdot 3^4 \cdot 5^2 \cdot 7^6 \cdot 19$
8 2^3		$\cong {}^2G_2(3)'$ $A_1(8)$ $2^3 \cdot 3^2 \cdot 7$	$A_2(8)$ $2^9 \cdot 3^2 \cdot 7^2 \cdot 73$	$\cong C_2(8)$ $B_2(8)$ $2^{12} \cdot 3^4 \cdot 5 \cdot 7^2 \cdot 13$	${}^2A_2(8)$ $2^9 \cdot 3^4 \cdot 7 \cdot 19$	$G_2(8)$ $2^{18} \cdot 3^5 \cdot 7^2 \cdot 19 \cdot 73$	${}^2B_2(8)$ $2^6 \cdot 5 \cdot 7 \cdot 13$		$A_3(8)$ $2^{18} \cdot 3^4 \cdot 5 \cdot 7^3 \cdot 13 \cdot 73$
9 3^2			$A_2(9)$ $2^7 \cdot 3^6 \cdot 5 \cdot 7 \cdot 13$	$B_2(9)$ $2^8 \cdot 3^8 \cdot 5^2 \cdot 41$	${}^2A_2(9)$ $2^5 \cdot 3^6 \cdot 5^2 \cdot 73$	$G_2(9)$ $2^8 \cdot 3^{12} \cdot 5^2 \cdot 7 \cdot 13 \cdot 73$			$A_3(9)$ $2^{10} \cdot 3^{12} \cdot 5^2 \cdot 7 \cdot 13 \cdot 41$
11 prime	C_{11} 11	$A_1(11)$ $2^2 \cdot 3 \cdot 5 \cdot 11$	$A_2(11)$ $2^4 \cdot 3 \cdot 5^2 \cdot 7 \cdot 11^3 \cdot 19$	$B_2(11)$ $2^6 \cdot 3^2 \cdot 5^2 \cdot 11^4 \cdot 61$	${}^2A_2(11)$ $2^5 \cdot 3^2 \cdot 5 \cdot 11^3 \cdot 37$	$G_2(11)$ $2^6 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11^6 \cdot 19 \cdot 37$			$A_3(11)$ $2^7 \cdot 3^2 \cdot 5^3 \cdot 7 \cdot 11^6 \cdot 19 \cdot 61$
13 prime	C_{13} 13	$A_1(13)$ $2^2 \cdot 3 \cdot 7 \cdot 13$	$A_2(13)$ $2^5 \cdot 3^2 \cdot 7 \cdot 13^3 \cdot 61$	$B_2(13)$ $2^6 \cdot 3^2 \cdot 5 \cdot 7^2 \cdot 13^4 \cdot 17$	${}^2A_2(13)$ $2^4 \cdot 3 \cdot 7^2 \cdot 13^3 \cdot 157$	$G_2(13)$ $2^6 \cdot 3^3 \cdot 7^2 \cdot 13^6 \cdot 61 \cdot 157$			$A_3(13)$ $2^7 \cdot 3^4 \cdot 5 \cdot 7^2 \cdot 13^6 \cdot 17 \cdot 61$
16 2^4		$A_1(16)$ $2^4 \cdot 3 \cdot 5 \cdot 17$	$A_2(16)$ $2^{12} \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 13 \cdot 17$	$\cong C_2(16)$ $B_2(16)$ $2^{16} \cdot 3^2 \cdot 5^2 \cdot 17^2 \cdot 257$	${}^2A_2(16)$ $2^{12} \cdot 3 \cdot 5 \cdot 17^2 \cdot 241$	$G_2(16)$ $2^{24} \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 13 \cdot 17^2 \cdot 241$			$A_3(16)$ $2^{24} \cdot 3^4 \cdot 5^3 \cdot 7 \cdot 13 \cdot 17^2 \cdot 257$
17 prime	C_{17} 17	$A_1(17)$ $2^4 \cdot 3^2 \cdot 17$	$A_2(17)$ $2^9 \cdot 3^2 \cdot 17^3 \cdot 307$	$B_2(17)$ $2^{10} \cdot 3^4 \cdot 5 \cdot 17^4 \cdot 29$	${}^2A_2(17)$ $2^6 \cdot 3^4 \cdot 7 \cdot 13 \cdot 17^3$	$G_2(17)$ $2^{10} \cdot 3^5 \cdot 7 \cdot 13 \cdot 17^6 \cdot 307$			$A_3(17)$ $2^{13} \cdot 3^4 \cdot 5 \cdot 17^6 \cdot 29 \cdot 307$
19 prime	C_{19} 19	$A_1(19)$ $2^2 \cdot 3^2 \cdot 5 \cdot 19$	$A_2(19)$ $2^4 \cdot 3^4 \cdot 5 \cdot 19^3 \cdot 127$	$B_2(19)$ $2^6 \cdot 3^4 \cdot 5^2 \cdot 19^4 \cdot 181$	${}^2A_2(19)$ $2^5 \cdot 3^2 \cdot 5^2 \cdot 7^3 \cdot 19^3$	$G_2(19)$ $2^6 \cdot 3^5 \cdot 5^2 \cdot 7^3 \cdot 19^6 \cdot 127$			$A_3(19)$ $2^7 \cdot 3^7 \cdot 5^2 \cdot 19^6 \cdot 127 \cdot 181$
23 prime	C_{23} 23	$A_1(23)$ $2^3 \cdot 3 \cdot 11 \cdot 23$	$A_2(23)$ $2^5 \cdot 3 \cdot 7 \cdot 11^2 \cdot 23^3 \cdot 79$	$B_2(23)$ $2^8 \cdot 3^2 \cdot 5 \cdot 11^2 \cdot 23^4 \cdot 53$	${}^2A_2(23)$ $2^7 \cdot 3^2 \cdot 11 \cdot 13^2 \cdot 23^3$	$G_2(23)$ $2^8 \cdot 3^3 \cdot 7 \cdot 11^2 \cdot 13^2 \cdot 23^6 \cdot 79$			$A_3(23)$ $2^9 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11^3 \cdot 23^6 \cdot 53 \cdot 79$
25 5^2		$A_1(25)$ $2^3 \cdot 3 \cdot 5^2 \cdot 13$	$A_2(25)$ $2^7 \cdot 3^2 \cdot 5^6 \cdot 7 \cdot 13 \cdot 31$	$B_2(25)$ $2^8 \cdot 3^2 \cdot 5^8 \cdot 13^2 \cdot 313$	${}^2A_2(25)$ $2^5 \cdot 3 \cdot 5^6 \cdot 13^2 \cdot 601$	$G_2(25)$ $2^8 \cdot 3^3 \cdot 5^{12} \cdot 7 \cdot 13^2 \cdot 31 \cdot 601$			$A_3(25)$ $2^{10} \cdot 3^4 \cdot 5^{12} \cdot 7 \cdot 13^2 \cdot 31 \cdot 313$
27 3^3		$A_1(27)$ $2^2 \cdot 3^3 \cdot 7 \cdot 13$	$A_2(27)$ $2^4 \cdot 3^9 \cdot 7 \cdot 13^2 \cdot 757$	$B_2(27)$ $2^6 \cdot 3^{12} \cdot 5 \cdot 7^2 \cdot 13^2 \cdot 73$	${}^2A_2(27)$ $2^5 \cdot 3^9 \cdot 7^2 \cdot 13 \cdot 19 \cdot 37$	$G_2(27)$ $2^6 \cdot 3^{18} \cdot 7^2 \cdot 13^2 \cdot 19 \cdot 37 \cdot 757$		${}^2G_2(27)$ $2^3 \cdot 3^9 \cdot 7 \cdot 13 \cdot 19 \cdot 37$	$A_3(27)$ $2^7 \cdot 3^{18} \cdot 5 \cdot 7^2 \cdot 13^3 \cdot 73 \cdot 757$
29 prime	C_{29} 29	$A_1(29)$ $2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 29$	$A_2(29)$ $2^5 \cdot 3 \cdot 5 \cdot 7^2 \cdot 13 \cdot 29^3 \cdot 67$	$B_2(29)$ $2^6 \cdot 3^2 \cdot 5^2 \cdot 7^2 \cdot 29^4 \cdot 421$	${}^2A_2(29)$ $2^4 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 29^3 \cdot 271$	$G_2(29)$ $2^6 \cdot 3^3 \cdot 5^2 \cdot 7^2 \cdot 13 \cdot 29^6 \cdot 67 \cdot 271$			$A_3(29)$ $2^7 \cdot 3^2 \cdot 5^2 \cdot 7^3 \cdot 13 \cdot 29^6 \cdot 67 \cdot 421$
31 prime	C_{31} 31	$A_1(31)$ $2^5 \cdot 3 \cdot 5 \cdot 31$	$A_2(31)$ $2^7 \cdot 3^2 \cdot 5^2 \cdot 31^3 \cdot 331$	$B_2(31)$ $2^{12} \cdot 3^2 \cdot 5^2 \cdot 13 \cdot 31^4 \cdot 37$	${}^2A_2(31)$ $2^{11} \cdot 3 \cdot 5 \cdot 7^2 \cdot 19 \cdot 31^3$	$G_2(31)$ $2^{12} \cdot 3^3 \cdot 5^2 \cdot 7^2 \cdot 19 \cdot 31^6 \cdot 331$			$A_3(31)$ $2^{13} \cdot 3^4 \cdot 5^3 \cdot 13 \cdot 31^6 \cdot 37 \cdot 331$
32 2^5		$A_1(32)$ $2^5 \cdot 3 \cdot 11 \cdot 31$	$A_2(32)$ $2^{15} \cdot 3 \cdot 7 \cdot 11 \cdot 31^2 \cdot 151$	$\cong C_2(32)$ $B_2(32)$ $2^{20} \cdot 3^2 \cdot 5^2 \cdot 11^2 \cdot 31^2 \cdot 41$	${}^2A_2(32)$ $2^{15} \cdot 3^2 \cdot 11^2 \cdot 31 \cdot 331$	$G_2(32)$ $2^{30} \cdot 3^3 \cdot 7 \cdot 11^2 \cdot 31^2 \cdot 151 \cdot 331$	${}^2B_2(32)$ $2^{10} \cdot 5^2 \cdot 31 \cdot 41$		$A_3(32)$ $2^{30} \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 11^2 \cdot 31^3 \cdot 41 \cdot 151$
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
q	p prime C_p p	$A_1(q)$ $q(q^2-1)/\gcd(2,q-1)$	$A_2(q)$ $q^3(q^3-1)(q^2-1)/\gcd(3,q-1)$	$B_2(q)$ $q^4(q^4-1)(q^2-1)/\gcd(2,q-1)$	${}^2A_2(q)$ $q^3(q^3+1)(q^2-1)/\gcd(3,q+1)$	$G_2(q)$ $q^6(q^6-1)(q^2-1)$	${}^2B_2(2^{2m+1})$ $2^{4m+2}(2^{4m+2}+1)(2^{2m+1}-1)$	${}^2G_2(3^{2m+1})$ $3^{6m+3}(3^{6m+3}+1)(3^{2m+1}-1)$	$A_3(q)$ $q^6(q^4-1)(q^3-1)(q^2-1)/\gcd(4,q-1)$

q = size of the field

Row **1** “F_1” (alternating groups):

Column	Cell
A_4	$\cong A_1(4) \cong A_1(5)$ A_5 $2^2 \cdot 3 \cdot 5$
A_5	$\cong A_1(9) \cong B_2(2)'$ A_6 $2^3 \cdot 3^2 \cdot 5$
A_6	A_7 $2^3 \cdot 3^2 \cdot 5 \cdot 7$
A_7	$\cong A_3(2)$ A_8 $2^6 \cdot 3^2 \cdot 5 \cdot 7$
A_8	A_9 $2^6 \cdot 3^4 \cdot 5 \cdot 7$
A_9	A_{10} $2^7 \cdot 3^4 \cdot 5^2 \cdot 7$
A_n	A_{n+1} $(n+1)!/2$

Row **q** (continued):

Column	Cell
B_3	$B_3(q)$ $q^9(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
C_3	$C_3(q)$ $q^9(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
${}^2A_3$	${}^2A_3(q)$ $q^6(q^4-1)(q^3+1)(q^2-1)/\gcd(4,q+1)$
A_4	$A_4(q)$ $q^{10}(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(5,q-1)$
B_4	$B_4(q)$ $q^{16}(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
C_4	$C_4(q)$ $q^{16}(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
D_4	$D_4(q)$ $q^{12}(q^6-1)(q^4-1)(q^4-1)(q^2-1)/\gcd(4,q^4-1)$
${}^2A_4$	${}^2A_4(q)$ $q^{10}(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(5,q+1)$
${}^2D_4$	${}^2D_4(q)$ $q^{12}(q^6-1)(q^4+1)(q^4-1)(q^2-1)/\gcd(4,q^4+1)$
F_4	$F_4(q)$ $q^{24}(q^{12}-1)(q^8-1)(q^6-1)(q^2-1)$
${}^3D_4$	${}^3D_4(q)$ $q^{12}(q^{12}-1)(q^6-1)(q^2-1)/(q^4-1)$
${}^2F_4$	${}^2F_4(2^{2m+1})$ $2^{24m+12}(2^{12m+6}+1)(2^{8m+4}-1)(2^{6m+3}+1)(2^{2m+1}-1)$
A_5	$A_5(q)$ $q^{15}(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(6,q-1)$
B_5	$B_5(q)$ $q^{25}(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
C_5	$C_5(q)$ $q^{25}(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
D_5	$D_5(q)$ $q^{20}(q^8-1)(q^6-1)(q^5-1)(q^4-1)(q^2-1)/\gcd(4,q^5-1)$
${}^2A_5$	${}^2A_5(q)$ $q^{15}(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(6,q+1)$
${}^2D_5$	${}^2D_5(q)$ $q^{20}(q^8-1)(q^6-1)(q^5+1)(q^4-1)(q^2-1)/\gcd(4,q^5+1)$
A_6	$A_6(q)$ $q^{21}(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(7,q-1)$
B_6	$B_6(q)$ $q^{36}(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
C_6	$C_6(q)$ $q^{36}(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
D_6	$D_6(q)$ $q^{30}(q^{10}-1)(q^8-1)(q^6-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^6-1)$
${}^2A_6$	${}^2A_6(q)$ $q^{21}(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(7,q+1)$
${}^2D_6$	${}^2D_6(q)$ $q^{30}(q^{10}-1)(q^8-1)(q^6+1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^6+1)$
E_6	$E_6(q)$ $q^{36}(q^{12}-1)(q^9-1)(q^8-1)(q^6-1)(q^5-1)(q^2-1)/\gcd(3,q-1)$
${}^2E_6$	${}^2E_6(q)$ $q^{36}(q^{12}-1)(q^9+1)(q^8-1)(q^6-1)(q^5+1)(q^2-1)/\gcd(3,q+1)$
A_7	$A_7(q)$ $q^{28}(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(8,q-1)$
B_7	$B_7(q)$ $q^{49}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
C_7	$C_7(q)$ $q^{49}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
D_7	$D_7(q)$ $q^{42}(q^{12}-1)(q^{10}-1)(q^8-1)(q^7-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^7-1)$
${}^2A_7$	${}^2A_7(q)$ $q^{28}(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(8,q+1)$
${}^2D_7$	${}^2D_7(q)$ $q^{42}(q^{12}-1)(q^{10}-1)(q^8-1)(q^7+1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^7+1)$
E_7	$E_7(q)$ $q^{63}(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^2-1)/\gcd(2,q-1)$
A_8	$A_8(q)$ $q^{36}(q^9-1)(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(9,q-1)$
B_8	$B_8(q)$ $q^{64}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
C_8	$C_8(q)$ $q^{64}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
D_8	$D_8(q)$ $q^{56}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^8-1)$
${}^2A_8$	${}^2A_8(q)$ $q^{36}(q^9+1)(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(9,q+1)$
${}^2D_8$	${}^2D_8(q)$ $q^{56}(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8+1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^8+1)$
E_8	$E_8(q)$ $q^{120}(q^{30}-1)(q^{24}-1)(q^{20}-1)(q^{18}-1)(q^{14}-1)(q^{12}-1)(q^8-1)(q^2-1)$
A_9	$A_9(q)$ $q^{45}(q^{10}-1)(q^9-1)(q^8-1)(q^7-1)(q^6-1)(q^5-1)(q^4-1)(q^3-1)(q^2-1)/\gcd(10,q-1)$
B_9	$B_9(q)$ $q^{81}(q^{18}-1)(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
C_9	$C_9(q)$ $q^{81}(q^{18}-1)(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(2,q-1)$
D_9	$D_9(q)$ $q^{72}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^9-1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^9-1)$
${}^2A_9$	${}^2A_9(q)$ $q^{45}(q^{10}-1)(q^9+1)(q^8-1)(q^7+1)(q^6-1)(q^5+1)(q^4-1)(q^3+1)(q^2-1)/\gcd(10,q+1)$
${}^2D_9$	${}^2D_9(q)$ $q^{72}(q^{16}-1)(q^{14}-1)(q^{12}-1)(q^{10}-1)(q^9+1)(q^8-1)(q^6-1)(q^4-1)(q^2-1)/\gcd(4,q^9+1)$
A_n	$A_n(q)$ $q^{n(n+1)/2}\prod_{i=2}^{n+1}(q^i-1)/\gcd(n+1,q-1)$
B_n	$B_n(q)$ $q^{n^2}\prod_{i=1}^{n}(q^{2i}-1)/\gcd(2,q-1)$
C_n	$C_n(q)$ $q^{n^2}\prod_{i=1}^{n}(q^{2i}-1)/\gcd(2,q-1)$
D_n	$D_n(q)$ $q^{n(n-1)}(q^n-1)\prod_{i=1}^{n-1}(q^{2i}-1)/\gcd(4,q^n-1)$
${}^2A_n$	${}^2A_n(q)$ $q^{n(n+1)/2}\prod_{i=2}^{n+1}(q^i-(-1)^i)/\gcd(n+1,q+1)$
${}^2D_n$	${}^2D_n(q)$ $q^{n(n-1)}(q^n+1)\prod_{i=1}^{n-1}(q^{2i}-1)/\gcd(4,q^n+1)$

Sporadic groups

Name	Group	Order
Mathieu	M_{11}	$2^4 \cdot 3^2 \cdot 5 \cdot 11$
Mathieu	M_{12}	$2^6 \cdot 3^3 \cdot 5 \cdot 11$
Mathieu	M_{22}	$2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$
Mathieu	M_{23}	$2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23$
Mathieu	M_{24}	$2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$
Janko	J_1	$2^3 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 19$
Hall–Janko	J_2	$2^7 \cdot 3^3 \cdot 5^2 \cdot 7$
Janko	J_3	$2^7 \cdot 3^5 \cdot 5 \cdot 17 \cdot 19$
Janko	J_4	$2^{21} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11^3 \cdot 23 \cdot 29 \cdot 31 \cdot 37 \cdot 43$
Higman–Sims	HS	$2^9 \cdot 3^2 \cdot 5^3 \cdot 7 \cdot 11$
McLaughlin	McL	$2^7 \cdot 3^6 \cdot 5^3 \cdot 7 \cdot 11$
Held	He	$2^{10} \cdot 3^3 \cdot 5^2 \cdot 7^3 \cdot 17$
Rudvalis	Ru	$2^{14} \cdot 3^3 \cdot 5^3 \cdot 7 \cdot 13 \cdot 29$
Suzuki	Suz	$2^{13} \cdot 3^7 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$
O’Nan	O’N	$2^9 \cdot 3^4 \cdot 5 \cdot 7^3 \cdot 11 \cdot 19 \cdot 31$
Conway	Co_1	$2^{21} \cdot 3^9 \cdot 5^4 \cdot 7^2 \cdot 11 \cdot 13 \cdot 23$
Conway	Co_2	$2^{18} \cdot 3^6 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$
Conway	Co_3	$2^{10} \cdot 3^7 \cdot 5^3 \cdot 7 \cdot 11 \cdot 23$
Fischer	Fi_{22}	$2^{17} \cdot 3^9 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13$
Fischer	Fi_{23}	$2^{18} \cdot 3^{13} \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 23$
Fischer	Fi_{24}'	$2^{21} \cdot 3^{16} \cdot 5^2 \cdot 7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 23 \cdot 29$
Harada–Norton	HN	$2^{14} \cdot 3^6 \cdot 5^6 \cdot 7 \cdot 11 \cdot 19$
Lyons	Ly	$2^8 \cdot 3^7 \cdot 5^6 \cdot 7 \cdot 11 \cdot 31 \cdot 37 \cdot 67$
Thompson	Th	$2^{15} \cdot 3^{10} \cdot 5^3 \cdot 7^2 \cdot 13 \cdot 19 \cdot 31$
Baby Monster	B	$2^{41} \cdot 3^{13} \cdot 5^6 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 31 \cdot 47$
Monster	M	$2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71$

CLASSIFICATION THEOREM

Every finite simple group is cyclic of prime order, alternating of degree n ≥ 5, a group of Lie type, or one of the 26 sporadic groups. This poster shows every one of them, each exactly once.

NOTES

- After the cyclic column the columns are grouped by rank: every family with a simple group of rank 1, then of rank 2, and so on.
- E_8 is the last family to appear, so from rank 9 onwards every block is the same six classical families — hence the ellipsis.
- Row q = 1: over “the field with one element” a group of Lie type degenerates to its Weyl group, and the cell $A_n(1)$ holds the alternating group A_{n+1}. Only type A is simple there, so that row is exactly the alternating groups.
- A cell carrying a parameter shows the generic order formula in place of a prime factorisation, and so does a cell whose factorisation is too long to print.
- $C_n(q) \cong B_n(q)$ for q even, so only the odd-q cells of the C columns are printed.
- ${}^2B_2(q)$ and ${}^2F_4(q)$ exist for $q = 2^{2m+1}$ and ${}^2G_2(q)$ for $q = 3^{2m+1}$ (m ≥ 1); their other rows are empty.
- ${}^2F_4(2)$ is not simple, but its derived subgroup — the Tits group — is, and is printed in that cell.
- The 26 sporadic groups belong to no family of Lie type; they are shown in their own strip.
- A symbol with a subscript and no field is a group over “F_1”: C_p is cyclic of order p and A_m is alternating of degree m. With a field, $A_n(q)$ is linear and $C_n(q)$ symplectic — the argument tells them apart.